MAST30027: Modern Applied Statistics

Assignment 1, 2026 — Graduate Analyst Client Portfolio

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Table of contents

Workplace scenario
Instructions
Question 1: Titanic survival from first principles (15 marks)
Question 2: Logistic regression with Antarctic penguins (6 marks)
Question 3: Gamma distribution (6 marks)

Due: 5:00 pm, Friday 21 August

This assignment is worth **7%** of your final mark. It has **27 marks**.

Workplace scenario

> ⓘ **Your role**
>
> You have just started work as a **Graduate Statistical Analyst** at **Meridian Insights**, a fictional analytics consultancy. Your first portfolio spans maritime history, Antarctic field science, and environmental monitoring. Each brief requires both technical justification and an analysis that a client can understand.

Your work will be read by two audiences:

1. **Technical reviewers**, who need to verify your mathematics, R code, assumptions, and calculations.
2. **Client decision-makers**, who are intelligent specialists in their own fields but are not familiar with generalised linear models, likelihood methods, link functions, or statistical jargon.

You will be given the QMD file containing these questions. Add your solutions directly to the same QMD file, placing each answer immediately below the corresponding question. Do not delete the questions or change their numbering. Replace the author placeholder in the YAML with your name and student ID, then render the completed file to HTML.

Although your answers are placed below the questions, the completed document should read as a professional workplace report for both audiences. Include the mathematics, R code, and output needed by the technical reviewer, and explain the method, results, and uncertainty in plain language for the client decision-maker. Organise each answer so that the explanation and supporting technical work are easy to connect.

Brief	Business context	Primary non-technical audience
1	Historical Titanic survival analysis	Maritime research managers
2	Antarctic penguin field-classification study	Field ecology managers
3	New York ozone model and methodology note	Environmental program managers

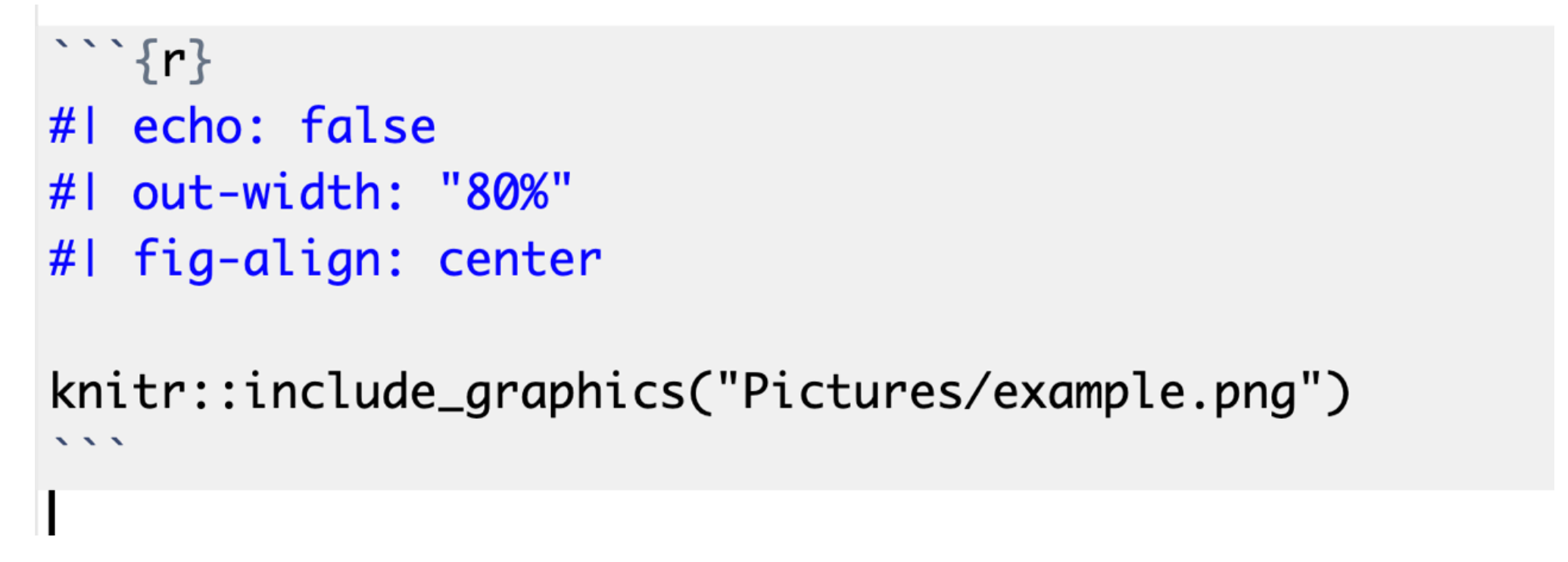
Instructions

- To get full marks, show your working, including the R commands and output you use, mathematical derivations, and clear explanations of how you reached your conclusions. Unsupported final answers may receive no marks.
- Add your solutions to the QMD file supplied with the assignment. The completed QMD must render successfully from beginning to end and must contain all package loading, data preparation, calculations, models, tables, and figures used in your answers. Do not rely on objects created interactively before rendering.
- All R code and all results produced by the code must be visible in the report. Do not fold or hide code or output. The supplied QMD sets `echo: true`; the equivalent option for an individual code chunk is `#| echo: true`.
- Create an RStudio Project for the assignment. Keep the supplied QMD file and the `images` folder inside the project folder, and open the project before working on or rendering the assignment.
- Unless a question states otherwise, do not use a function that directly performs the calculation you have been asked to derive.
- Use a two-sided significance level of 5% and 95% confidence intervals unless stated otherwise. Label all tables and figures clearly.
- Your answers must be clearly numbered and presented in the same order as the questions.

Mathematical derivations

You can type mathematical derivations using LaTeX or complete the derivations clearly by hand and include a picture of them in your report. If using handwritten work, scan or photograph it, crop it closely, and save it as a PNG image. Store the image in an `images` folder beside your QMD and insert it at the appropriate point in the report. The image must be upright, high contrast, legible at normal PDF magnification, and labelled with the relevant question number.

You can include an image with `knitr::include_graphics()`, as illustrated below. Replace the example file path with the relative path to your own image.



~~~
```{r}
#| echo: false
#| out-width: "80%"
#| fig-align: center

knitr::include_graphics("Pictures/example.png")
```
~~~

Example of including a PNG image in a Quarto document.

Handwritten work is part of the report: it must appear in the rendered HTML and in the final PDF. Separate image files are not submitted.

Submission

How to create the PDF from your HTML report

1. Render the QMD file to HTML and open the HTML file in a web browser such as Chrome, Edge, Firefox or Safari.
2. Open the browser's Print window by pressing **Ctrl+P** on Windows or **Command+P** on macOS.
3. Select **Save as PDF**, **Microsoft Print to PDF**, or **PDF > Save as PDF**, depending on your browser and operating system.
4. Check the print preview. Use an appropriate page orientation and scale so that code, tables and graphs are not cut off. Turn off browser headers and footers if they obscure the report.
5. Save the file as a PDF named `studentID_assignment1.pdf`. Open the saved PDF and inspect every page before submitting it.

Check that all code, output, equations, tables, figures, and handwritten work are readable and are not cut off. Submit this single PDF through the LMS. Do not submit the QMD or HTML files. It is your responsibility to upload the correct PDF on time.

Question 1: Titanic survival from first principles (15 marks)

Client brief: historical maritime survival analysis

A maritime research organisation is preparing an evidence-based exhibition about survival on the Titanic. Its managers want a transparent analysis of how sex and age group were associated with survival among third-class passengers. Because the case is historically sensitive, distinguish statistical association from explanations of individual behaviour or policy.

The built-in `Titanic` dataset cross-tabulates 2,201 people by class, sex, age group, and survival. It is based on records originally collected by the British Board of Trade. The data are in the `datasets` package, which is included with R; no additional package installation is required. Load and prepare the data using:

```r
data(Titanic)
titanic_long <- as.data.frame(Titanic)
titanic_grouped <- reshape(
  titanic_long,
  idvar = c("Class", "Sex", "Age"),
  timevar = "Survived",
  direction = "wide"
)
titanic_grouped$Total <-
  titanic_grouped$Freq.No + titanic_grouped$Freq.Yes
titanic_grouped <- subset(
  titanic_grouped,
  Class == "3rd" & Total > 0
)
titanic_grouped$female <- as.integer(titanic_grouped$Sex == "Female")
titanic_grouped$adult <- as.integer(titanic_grouped$Age == "Adult")
str(titanic_grouped)
```

Fit a binomial regression model to the survival data for third-class passengers, using a *probit* link. For group i, let Y_i be `Freq.Yes`, let m_i be `Total`, and use the model

$$Y_i \sim \text{Binomial}(m_i, p_i), \qquad \Phi^{-1}(p_i) = \beta_0 + \beta_1 \texttt{female}_i + \beta_2 \texttt{adult}_i,$$

where `female` equals 1 for females and 0 for males, and `adult` equals 1 for adults and 0 for children. You must use R, but for parts (a)–(d) you must not use `glm()`, `glm.fit()`, or another function that fits a generalised linear model. Work from first principles. You may use a general-purpose optimiser such as `optim()`.

(a) Maximum likelihood estimation (3 marks). Derive the log-likelihood, up to an additive constant, and compute the maximum likelihood estimates of β_0, β_1, and β_2. Show the R function used to evaluate the log-likelihood and state how you checked convergence.

(b) Fisher information and confidence intervals (6 marks). Compute 95% confidence intervals for the three parameter estimates. Show how you derived the Fisher information matrix and how you evaluated it in R.

(c) Likelihood-ratio test (3 marks). Perform a likelihood-ratio test for the significance of the `female` coefficient after accounting for age group. State the null and alternative hypotheses, test statistic, reference distribution, degrees of freedom, p-value, and conclusion in context.

(d) Question from the maritime research manager (3 marks).

> "For a third-class adult female in these historical records, what survival probability does the model estimate? How certain are we about this estimate?"

Answer the manager's question by computing the estimated survival probability and a 95% confidence interval. Form the confidence interval on the linear-predictor scale and transform its endpoints using Φ. Show the calculation and interpret the result in context.

Question 2: Logistic regression with Antarctic penguins (6 marks)

Client brief: field classification from body measurements

An Antarctic field-research team wants to understand the association between body mass and the recorded sex of adult penguins. The goal is to describe a pattern in these observations, not to claim that changing body mass would change an animal's sex.

The `penguins` data in the `palmerpenguins` package contain measurements for 344 Adélie, Chinstrap, and Gentoo penguins observed near Palmer Station, Antarctica, from 2007 to 2009. If the package is not already installed, install it once by running `install.packages("palmerpenguins")` in the R console; do not include the installation command in the report. Load the package and prepare the data using:

```r
library(palmerpenguins)
data(penguins)
penguin_analysis <- na.omit(
  penguins[c("sex", "body_mass_g")]
)
penguin_analysis$male <- as.integer(penguin_analysis$sex == "male")
str(penguin_analysis)
```

You may use `glm()` in this question.

The field ecology manager asks:

> "According to these data, how do the odds of a male record change when body mass increases by 500 g? How certain are we about the estimated change?"

(a) Change in log-odds (3 marks). Fit a binomial regression with a logit link, using `male` as the response and `body_mass_g` as the predictor. Starting from the model equation, derive the relationship between the body-mass coefficient and the change in log-odds associated with a 500 g increase. Estimate this change.

(b) Confidence interval and interpretation (3 marks). Compute a 95% confidence interval for the change in log-odds from part (a). Exponentiate the estimate and interval, and use the resulting odds ratio to answer the manager's question in context. State one limitation of this simple model.

Question 3: Gamma distribution (6 marks)

Client brief: methodology note

Your technical reviewer requires the exponential-family derivation for gamma distribution.

The gamma distribution with shape $\nu > 0$ and rate $\lambda > 0$ has density

$$f(y; \nu, \lambda) = \frac{\lambda^{\nu}}{\Gamma(\nu)} y^{\nu-1} e^{-\lambda y}, \qquad y > 0.$$

(a) Exponential-family form (3 marks). Rewrite this density in exponential-dispersion-family form,

$$f(y; \theta, \phi) = \exp\left\{ \frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi) \right\}.$$

Identify the natural parameter θ, dispersion parameter ϕ, and valid expressions for $a(\phi)$, $b(\theta)$, and $c(y, \phi)$. State the parameter space.

(b) Canonical link and variance function (3 marks). Using your exponential-family representation, derive $E(Y)$ and $\text{Var}(Y)$. Hence obtain the canonical link and the variance function $V(\mu)$ defined by $\text{Var}(Y) = a(\phi)V(\mu)$.